



FIRST SEMESTER 2025-2026
Course Handout Part II

Date: 02/08/2026

In addition to Part-I (General Handout for all courses appended to the Time Table) this portion gives further specific details regarding the course.

Course No. : CHE G552
Course Title : Advanced Transport Phenomena | ATP
Instructor-in-charge : SAIKAT BHATTACHARJEE

- 1. Course Description:** Viscosity, thermal conductivity and diffusivity, Shell momentum and energy balance, equations of change for isothermal and non-isothermal systems, Concentration distribution in solids and laminar flows, momentum, thermal and concentration boundary layers near walls, origin of turbulence, length scales in turbulent flows, Reynolds (RANS) equations, estimates of Reynolds stress (k -epsilon and k -omega type models), turbulent shear flow near a wall, turbulent flow in pipes and channels, turbulent heat transfer, Introduction to large eddy simulations models, rheology and material functions, non-Newtonian viscosity and generalized Newtonian models, Linear and non-linear visco-elasticity, radiation heat transfer, multi-component systems, Coupled heat and mass transfer, evaporation, boiling and condensation, chemical reactions, *Special topics*: Flow through porous media, compressible flows, multiphase flow, Transport phenomena in biochemical systems.
- 2. Scope & Objective:** Transport phenomena is one of the basic subjects of chemical engineering and includes study of fluid dynamics, heat and mass transfer. The governing equations for the aforementioned transport processes are analogous in nature as the underlying molecular mechanisms are very closely related. The objective of this course is to cover advance topics in this area like the transport of momentum, heat and mass in turbulent flow, creeping flows, flow through porous media, flow over flat plates and curved surfaces, interphase transport, etc. A balanced overview and fundamental equations for transport processes would be provided along with illustrations regarding solving relevant problems. Further, some special cases which are frequently encountered would also be covered.

3. Text Book (TB | T):

T1 Bird, R. Byron; Stewart, Warren E and Lightfoot, Edwin N | **BSL** | “Transport Phenomena”, John Wiley & Sons, 2nd Ed, 2002 [*or the latest available!*].

T2 Plawsky Joel L, “Transport Phenomena Fundamentals”, CRC Press, 3rd Ed, 2014.

4. Reference Books (RB | R):

R1 Leal, L. Gary “Advanced Transport Phenomena”, Cambridge University Press, 2007.

R2 Larson, Ronald G “The Structure and Rheology of Complex Fluids”, Oxford University Press, 1999.

R3 Rubinstein, Michael and Colby, Ralph H “Polymer Physics”, Oxford University Press, 2003.

R4 Deen, William M “Analysis of Transport Phenomena”, Oxford University Press [Indian Edition], 2013 (Reprint!).

R5 Slattery, John C “Advanced Transport Phenomena”, Cambridge University Press, 1999.

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5. Course Plan:

Module (M) / Lecture No	Learning Objectives #	Topics to be covered	Reference Chap / Sec (Book)
1-2	Introduction to the course	Scope, Objectives, Methodology, Importance, Preliminary assumptions, Foundations and concepts related to transport phenomena, Viscosity, thermal conductivity and diffusivity	Ch. 0, 1, 9, 17 (T1) Ch. 1-3 (T2)
3-5	Review of Shell Momentum balances, Reynolds Transport Theorem, Equations of change for isothermal systems	Shell Momentum Balances and Boundary Conditions, The Equations of Change in terms of the Substantial Derivative, Use of the Equations of Change to Solve Flow Problems	Ch. 2-3 (T1)
5-6	Review of Shell Energy balances, Equations of change for non-isothermal systems	Shell Energy Balances and boundary conditions, Free and forced convection, Use of the Equations of Change to Solve Steady-State Problems	Ch. 10-11 (T1)
7-8	Review of Concentration Distributions in Solids and in Laminar Flow	Shell mass balances, Boundary conditions, Diffusion through a stagnant gas film, with a chemical reaction, Diffusion and chemical reaction inside a porous catalyst, Multicomponent equations of change, multicomponent fluxes, use of the equations of change for mixtures	Ch. 18-19 (T1)
9-10	Review of boundary layer concepts, Convective Transport on a flat plate, Convective Transport over systems with curvature	Laminar hydrodynamic, thermal and mass transfer boundary layers, flow over cylinders, spheres; flow in tubes	Ch. 12-13 (T2)
11-15	Velocity Distributions in Turbulent Flow	Basic features of turbulence, Time-Smoothed Equations of Change for Incompressible Fluids, The Time-Smoothed Velocity Profile near a Wall, Empirical Expressions for the Turbulent Momentum Flux, Turbulent flow in pipes and channels, Eddy diffusivity models	Ch. 5 (T1) Ch. 14 (T2) Ch. 13 (R4)
16-17	Temperature distributions in Turbulent flow	Time-Smoothed Equations of Change for Incompressible Non-isothermal Flow, The Time-Smoothed Temperature Profile near a Wall, Empirical Expressions for the Turbulent Heat Flux, Temperature	Ch. 13 (T1)



		Distribution for Turbulent Flow in Tubes and jets	
18-19	Concentration Distributions in Turbulent Flow	Time-smoothing of the equation of continuity, Enhancement of mass transfer by a first-order reaction in turbulent flow, Turbulent mixing and turbulent flow with second-order reaction	Ch. 21 (T1)
20-22	Interphase transport in non-isothermal systems and mixtures	Heat Transfer Coefficients for Forced Convection, Free and Mixed Convection, Heat Transfer Coefficients for Condensation of Pure Vapors on Solid Surfaces, Definition of transfer coefficients in two phases, Effects of interfacial forces on heat and mass transfer	Ch. 6, 14, 22 (T1)
23-27	Coupled heat and mass transfer	Coupled diffusion and reaction, coupled heat and momentum transport	Ch. 5 (T2) Ch. 14 (R4)
28-30	Energy transport by radiation	Absorption and emission at solid surfaces, Planck's distribution law, Wien's displacement law, and the Stefan-Boltzmann law, Radiation between non-black bodies at different temperatures	Ch. 16 (T1) Ch. 15 (T2)
31-33	Rheology of complex fluids, Linear and non-linear viscoelasticity	Fundamentals of rheology, types of complex fluids, origins of Non-Newtonian behavior	Ch. 1, 3 (R2)
34-37	Behavior of polymeric melts	Non-Newtonian viscosity and generalized Newtonian models, elasticity and linear viscoelastic models, un-entangled and entangled polymer melts	Ch. 8 (T1) Ch. 8-9 (R3)
38-41	Special topics: Flow through porous media, multiphase flow, Hydrodynamic Stability	Heat and mass transfer using extended surfaces, diffusion and reaction in catalyst pellets Multidimensional effects, potential functions and fields	Ch. 8-9 (T2) Ch. 12 (R1)

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6. Evaluation Scheme*

Component	Duration	Weightage (%)	Date and Time	Nature of Component
Mid-Semester test	90 min	20	To be announced in class	OB
Comprehensive Examination	180 min	30	To be announced in class	CB
Lecture Quiz (3 out of 4)	30 min	10	To be announced in class	CB
Assignment (2)	-	20	-	OB (Take home)
Seminar presentation	-	10	-	-
Writing of a short review article on the topic chosen and discussed with the IC	-	10	-	-

* If any change in this/rest of the handout: it will be communicated in Classes – pls follow them.

** CB = Close Book | OB = Open Book: **Only Text Book(s) will be permitted. Nothing Else!** / Also implies "Take Home!"

A judicious blend of these components will spread over the semester. More details to be shared.

7. **Chamber consultation hour** will be announced in the class.

8. The **notices**, if any, concerning the course, will be shared.

9. **Make-up** will be granted for genuine medical cases only. Proper justification from the warden of the respective bhavan, certificate from authenticated doctor, and from the in-campus medical center must accompany make-up application (*only prescription or vouchers for medicines will not be sufficient*). Prior permission of IC is compulsory. **The student must refer to the procedure prescribed by AUGSD for applying for Make-up.**

make up for project/seminars/assignments/class test/class participation.

Saikat Bhattacharya

Instructor-in-charge
CHE G552